Radio Frequency Engineering

Exercise-5 Laboratory report

SS2026

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STE25
July 31, 2026

Contents

1 Part 1: Become familiar with the Spectrum Analyzer

1.1 Task 1: Analyze the VCO ZOS-100 Oscillator

1.1.1 Task Description

The objective of this task is to analyze the performance of the VCO ZOS-100+ Oscillator. This involves determining its frequency range, measuring the VCO tuning sensitivity (in Hz/V), identifying the minimum and maximum frequencies within the valid control voltage range, and recording the power levels delivered at its outputs.

1.1.2 Schematic

Following schematic shows the oscillator connected to a spectrum analyser and a DC power supply.

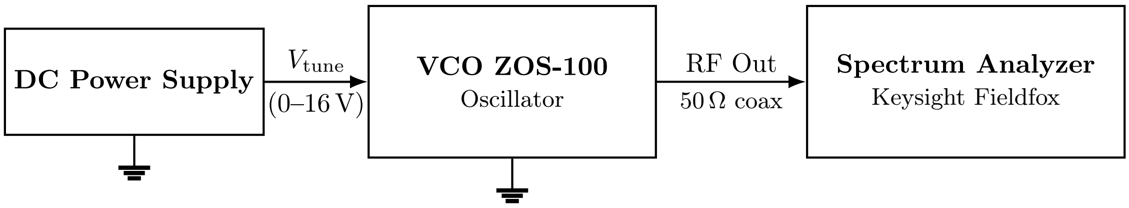

Figure 1.1: Measurement setup with a spectrum analyser for the ZOS-100 oscillator.

1.1.3 Formulas and Calculations

To determine the tuning sensitivity of the oscillator, the following formula is used:

$$K_v = \frac{\Delta f}{\Delta V} = \frac{f_{\max} - f_{\min}}{V_{\max} - V_{\min}} \tag{1.1}$$

1.1.4 Measurement results

The following table summarizes the measured frequencies and power levels from the spectrum analyser.

Adjusted	**Measured**	
Control Voltage (V_{tune}) / V	**Frequency (f) / MHz**	**Output Power / dBm**
[Insert Voltage]	54.750	9.962
[Insert Voltage]	75.000	9.368
10.0	84.450	9.005

Table 1.1: Frequency and Output Power of the VCO ZOS-100+

Notes:

- Measurement at 54.750 MHz was recorded from the “aux out”.
- As mandated by the report guidelines, ensure that the columns clearly separate adjusted, measured, and calculated values.

1.1.5 Curves & diagrams

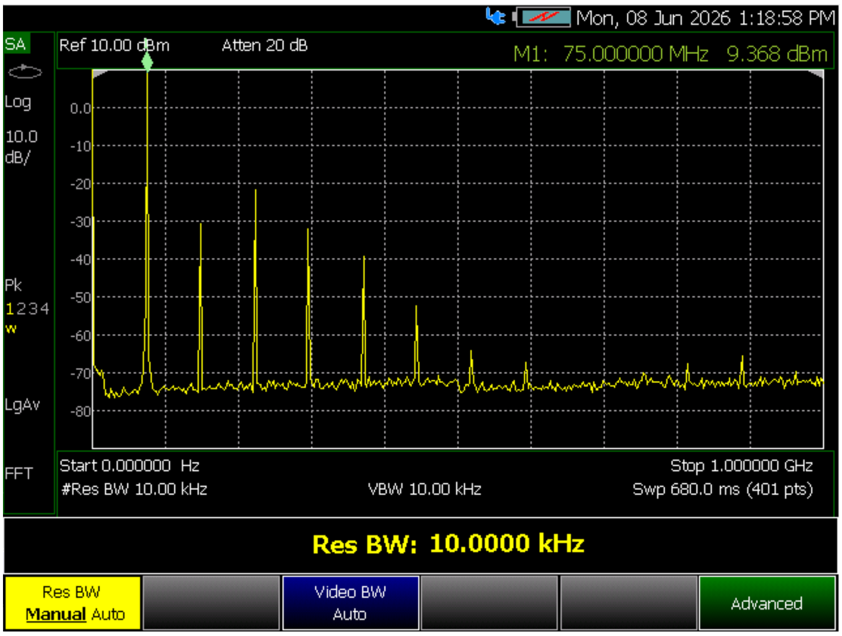

Figure 1.2: Broadband measurement of the VCO ZOS-100+ output showing the fundamental frequency peak at 75.000 MHz (9.368 dBm) along with multiple visible harmonics.

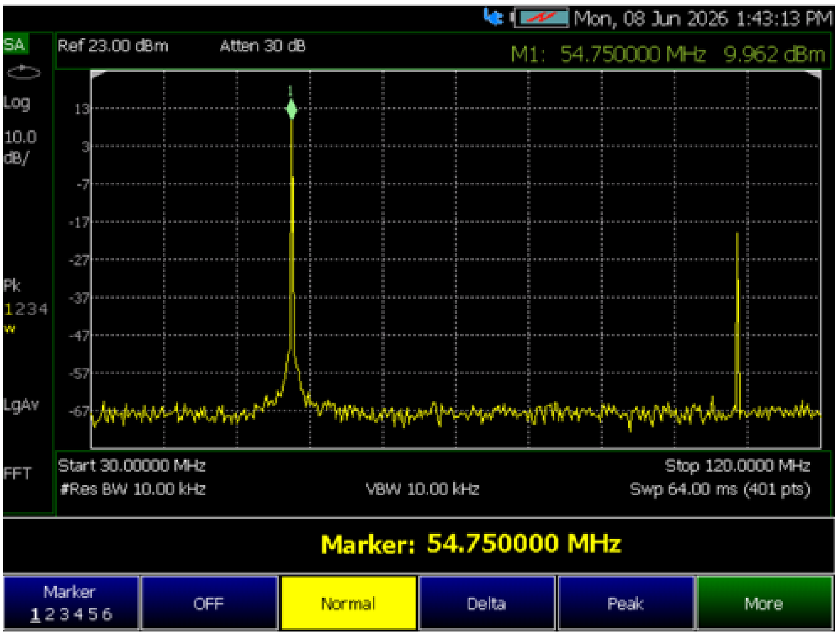

Figure 1.3: Spectrum analyzer measurement of the VCO “aux out” displaying a fundamental frequency of 54.750 MHz with an output power of 9.962 dBm.

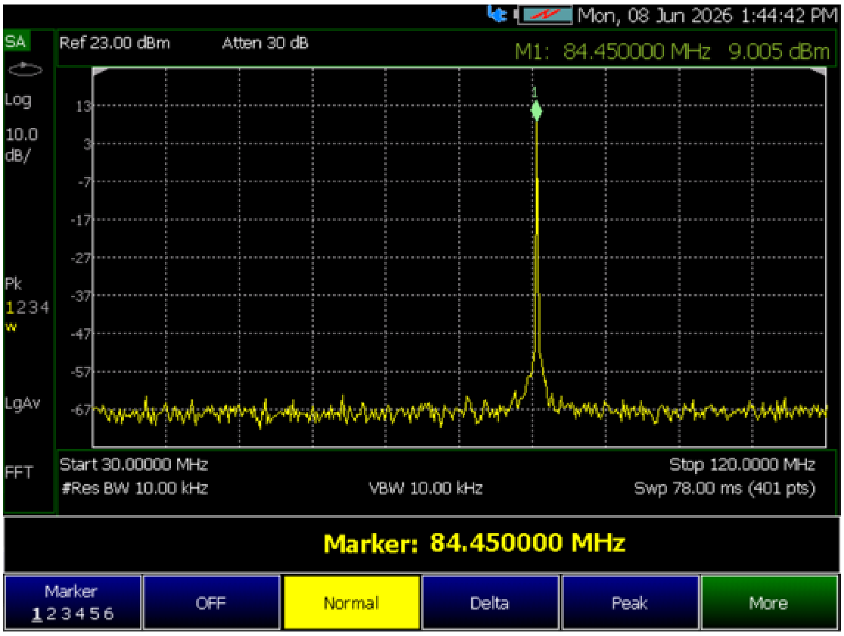

Figure 1.4: Spectrum analyzer measurement of the VCO at a control voltage of 10V, showing a fundamental frequency of 84.450 MHz with an output power of 9.005 dBm.

1.1.6 Short discussion of measurement results

The results demonstrate that the fundamental frequency of the VCO ZOS-100+ shifts according to the applied control voltage, yielding a measured tuning range from 54.750 MHz up to 84.450 MHz (at a 10V control voltage). Across this tuning range, the oscillator maintains a relatively stable output power, fluctuating only slightly between 9.005 dBm and 9.962 dBm. Additionally, the broadband measurement clearly displays multiple expected harmonic frequencies occurring alongside the fundamental peak.

1.2 Task 2: Analyze another RF Oscillator

1.2.1 Task Description

The objective of this task is to analyze the spectrum of another selected RF oscillator from the hardware inventory. This includes determining the frequencies and power levels of the most powerful spurious emissions (spurii) and evaluating whether these frequencies are harmonics of the fundamental frequency. (Note: The bonus measurement for phase noise was not conducted and is excluded from this report).

1.2.2 Schematic

Following schematic shows the oscillator connected to a spectrum analyser and a DC power supply.

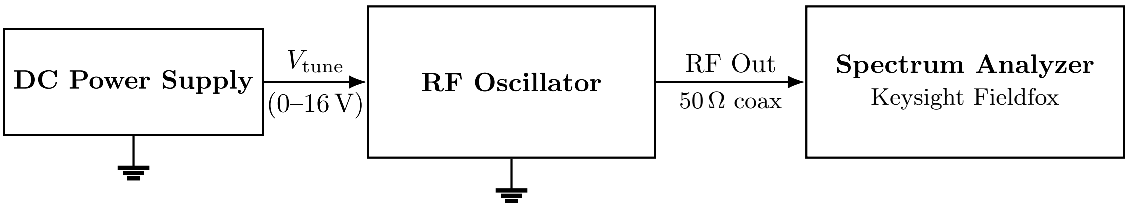

Figure 1.5: Measurement setup with a spectrum analyser for the ZOS-100 oscillator.

1.2.3 Formulas and Calculations

To determine if a spurious emission is a harmonic, its frequency must be an integer multiple of the fundamental frequency (f_1):

$$f_n = n \cdot f_1 \tag{1.2}$$

where n is an integer $(2, 3, 4, \dots)$.

1.2.4 Measurement results

The provided data explicitly identifies two specific frequency peaks. Because the exact identity of the fundamental frequency was not explicitly recorded, the harmonic calculation (n) cannot be definitively confirmed.

Adjusted	**Measured**	
Control Voltage / V	**Frequency (f) / MHz**	**Output Power / dBm**
[Unknown]	342.500	−27.50
[Unknown]	867.500	−33.51

Table 1.2: Measured frequencies and power levels of the unknown RF Oscillator

1.2.5 Curves & diagrams

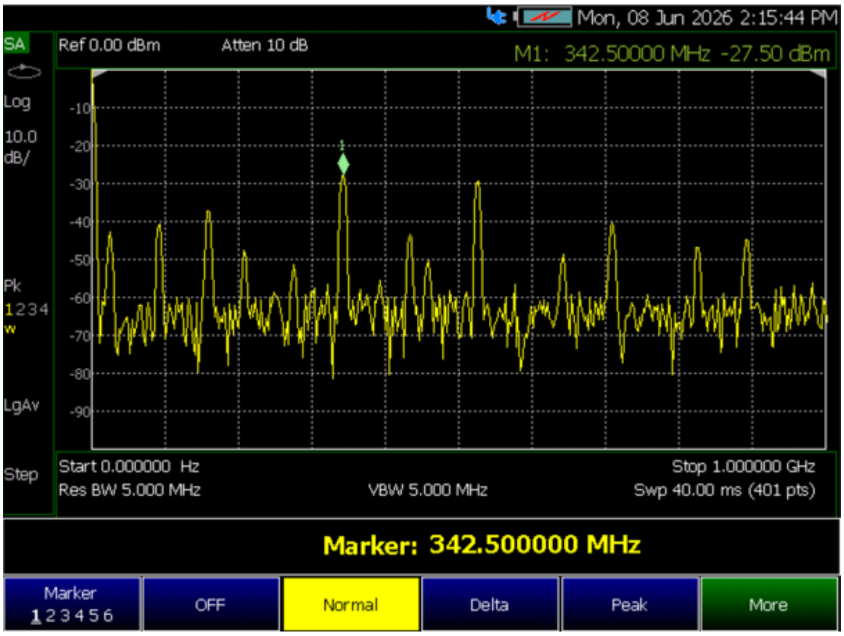

Figure 1.6: Spectrum analyzer measurement of the unknown RF oscillator from 0 Hz to 1 GHz, displaying multiple frequency peaks with a specific marker placed at a spurious emission of 342.500 MHz (-27.50 dBm).

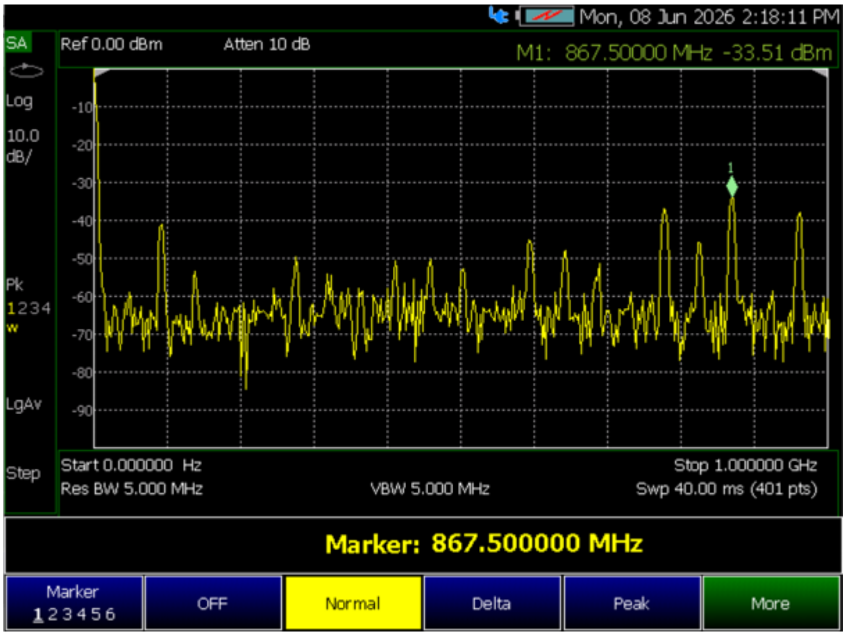

Figure 1.7: Spectrum analyzer measurement of the unknown RF oscillator, highlighting a higher-frequency spurious emission at 867.500 MHz with a measured power level of -33.51 dBm.

1.2.6 Short discussion of measurement results

The wideband spectrum of the unknown RF oscillator displays multiple spurious emissions. Two specific frequency peaks were measured at 342.500 MHz and 867.500 MHz. Because 867.500 MHz is not an integer multiple of 342.500 MHz, these two emissions are not direct harmonics of each other. A full harmonic analysis cannot be completed because the true fundamental frequency of the oscillator was not explicitly documented during the measurement.

1.3 Task 3: Analyze the function of a double balanced RF Mixer

1.3.1 Task Description

The objective of this task is to analyze the performance of a double-balanced RF mixer. This involves applying Local Oscillator (LO) and Intermediate Frequency (IF) signals to the mixer, and observing the wideband output spectrum to determine the wanted and unwanted frequency products.

1.3.2 Schematic

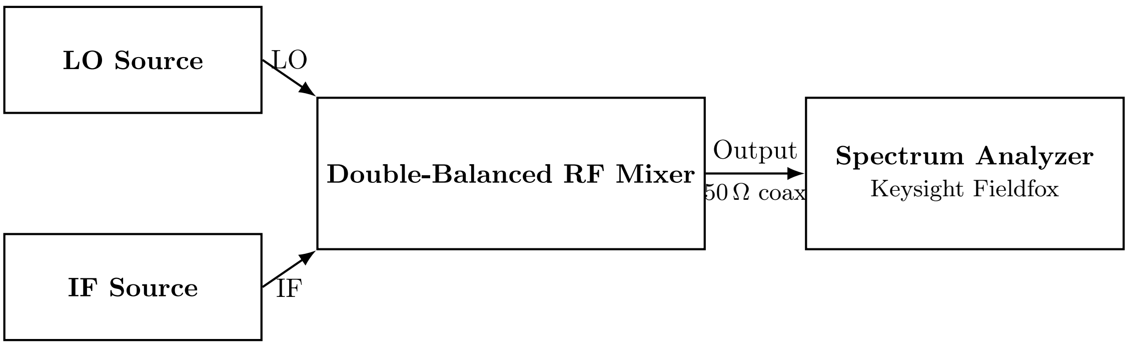

Figure 1.8: Measurement setup for the double-balanced RF mixer.

1.3.3 Formulas and Calculations

For a standard RF mixer, the output frequency products (f_{RF}) are mathematically defined by the sum and difference of the Local Oscillator (f_{LO}) and Intermediate Frequency (f_{IF}) input signals:

$$f_{\mathrm{RF}} = |f_{\mathrm{LO}} \pm f_{\mathrm{IF}}| \tag{1.3}$$

1.3.4 Measurement results

The data file contains a single prominent measurement for the mixer output. Because the applied LO and IF frequencies are missing, the “adjusted” and “calculated” columns currently contain placeholders.

Adjusted	Measured	
Input Frequencies (f_{LO}, f_{IF}) / MHz	**Output Frequency (f_{RF}) / MHz**	**Output Power / dBm**
[Unknown]	63.075	−11.78

Table 1.3: Measured output frequency and power level of the RF Mixer

1.3.5 Curves & diagrams

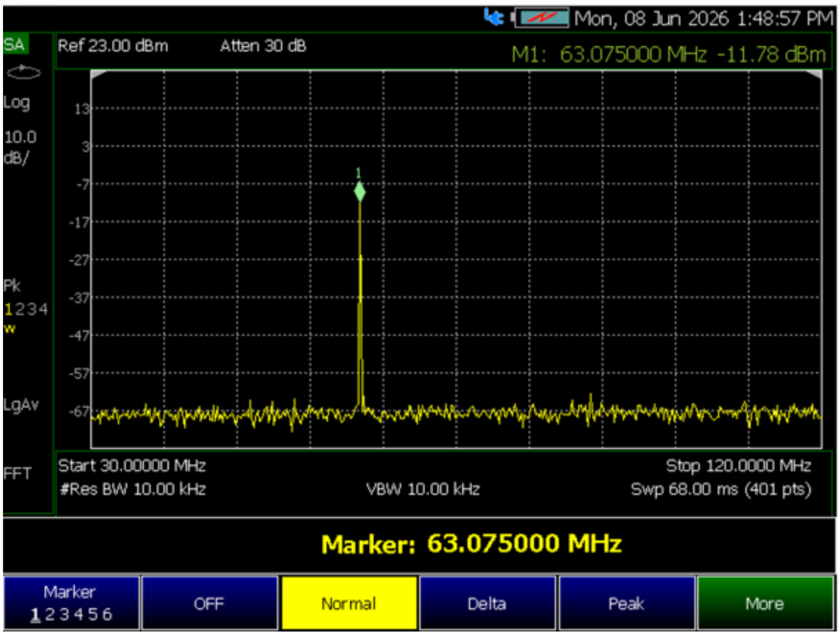

Figure 1.9: Spectrum analyzer measurement of the double-balanced RF mixer output, showing a single specific marker placed at a prominent frequency product of 63.075 MHz with a power level of -11.78 dBm.

1.3.6 Short discussion of measurement results

The spectrum analyzer measurement of the double-balanced RF mixer reveals a distinct frequency product at 63.075 MHz with a measured power level of −11.78 dBm. A full evaluation of the wanted and unwanted frequency products cannot be definitively concluded based on this single plot, as the specific f_{LO} and f_{IF} input frequencies applied during the experiment were not explicitly documented in the recorded dataset.

2 Part 2: Become familiar with the Network Analyzer

2.1 Task 1: Analyze given Terminations and Coaxial Cables with different loads

2.1.1 Task Description

The primary objective of this task is to utilize a Vector Network Analyzer (VNA), specifically the Keysight Fieldfox, to determine the performance parameters of various 50 Ω and 75 Ω terminations, as well as to analyze reflection coefficients when combining terminations using a BNC Coax-Tee and a resistive Power Divider. A crucial part of this exercise involves comparing measurements when the VNA is calibrated with and without a cable attached, and observing the effects of open and short calibrations.

2.1.2 Schematic

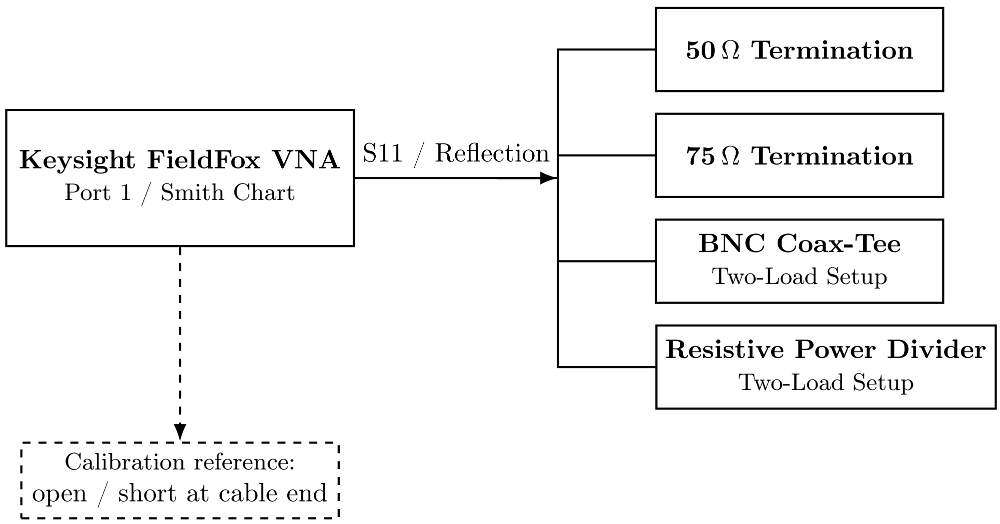

Figure 2.1: VNA measurement setup for terminations, coaxial cable, and combined loads.

2.1.3 Formulas and Calculations

The measurements rely on the fundamental relationship between the load impedance (Z) and the characteristic impedance of the system (Z_w, which is 50 Ω in this RF system). The reflection coefficient (ρ) at the load is defined as:

$$\rho = \frac{Z - Z_w}{Z + Z_w} \tag{2.1}$$

The Vector Network Analyzer displays these results on a Smith chart, which uniquely maps the complex normalized impedance plane ($z = Z/Z_w = r + jx$) to the complex reflection coefficient (ρ) plane.

2.1.4 Measurement results

The following table summarizes the real and imaginary impedance components extracted from the VNA Smith chart markers after calibration. Standard theoretical values are placed in the calculated column for comparison.

2.1.5 Curves & diagrams

Adjusted	**Measured**		**Calculated**
Termination Type	**Frequency / MHz**	**Measured Impedance (Z) / Ω**	**Theoretical Ideal (Z) / Ω**
$75\,\Omega$ Load (Post-Cal)	50.875	$74.3 + j0.3$	$75 + j0$
Short Load (Post-Cal)	50.875	$-0.3 + j0.7$	$0 + j0$
"One by 2" Termination	50.875	$50.2 + j1.3$	$50 + j0$
"Two by 2" Termination	50.875	$24.8 + j1.8$	$25 + j0$
Load Short (with cable)	48.923	$1.3\text{k} - j1.1\text{k}$	–

Table 2.1: Measured Impedances of Various Terminations on the VNA

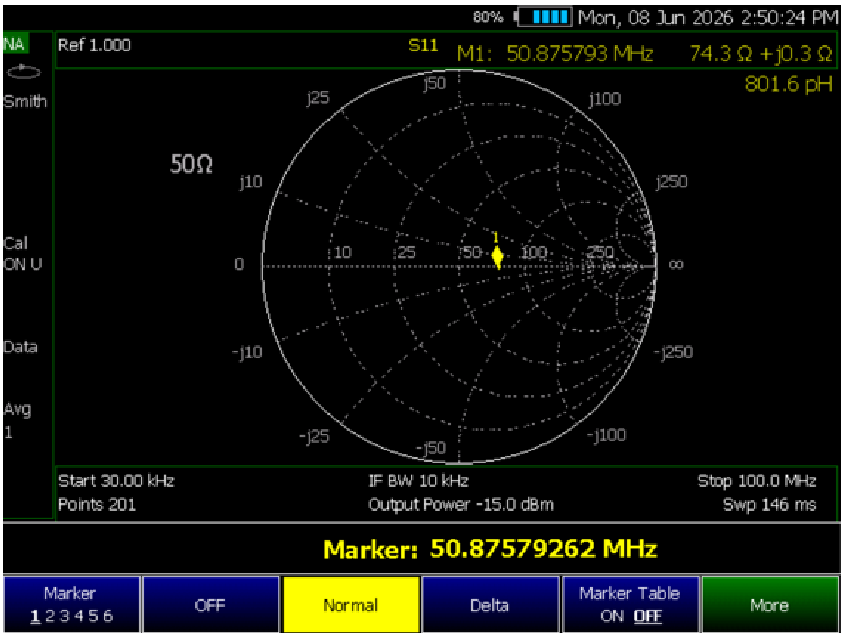

Figure 2.2: Smith chart measurement of a $75\,\Omega$ termination after VNA calibration, showing a marker at 50.875 MHz with a measured impedance of $74.3\,\Omega + j0.3\,\Omega$.

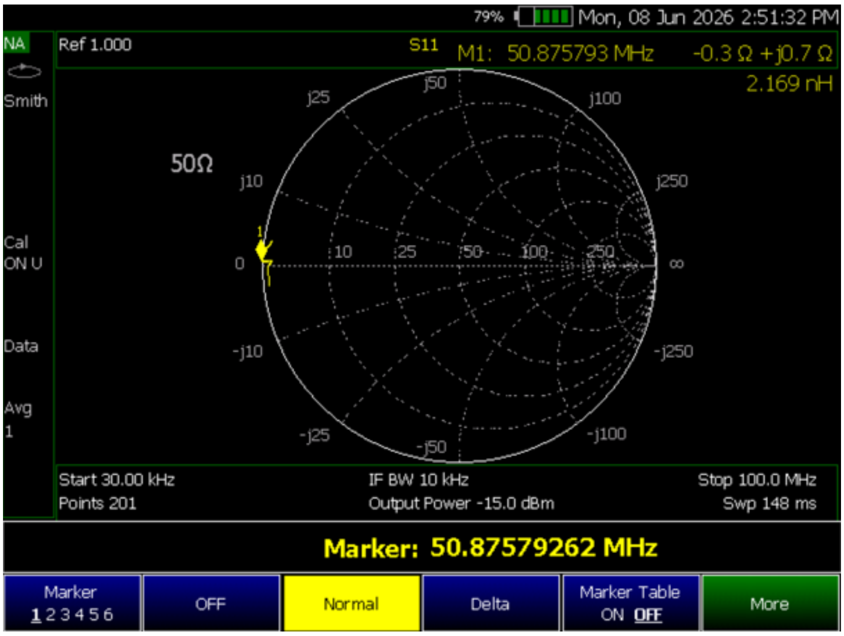

Figure 2.3: Smith chart measurement of a short circuit termination after calibration, verifying the short with a reading of $-0.3\,\Omega + j0.7\,\Omega$ at 50.875 MHz.

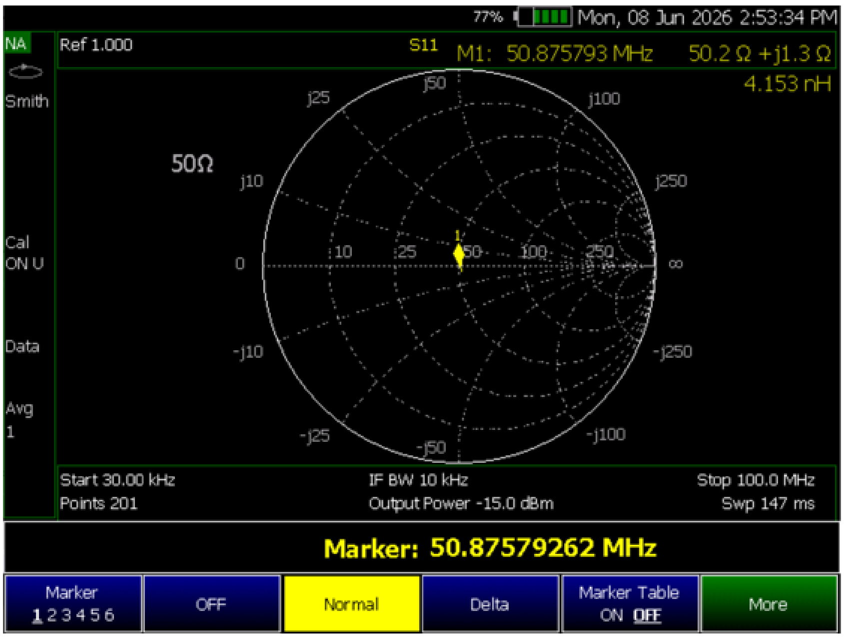

Figure 2.4: Smith chart measurement of the "One by 2" termination setup, displaying an impedance of $50.2\,\Omega + j1.3\,\Omega$ at 50.875 MHz.

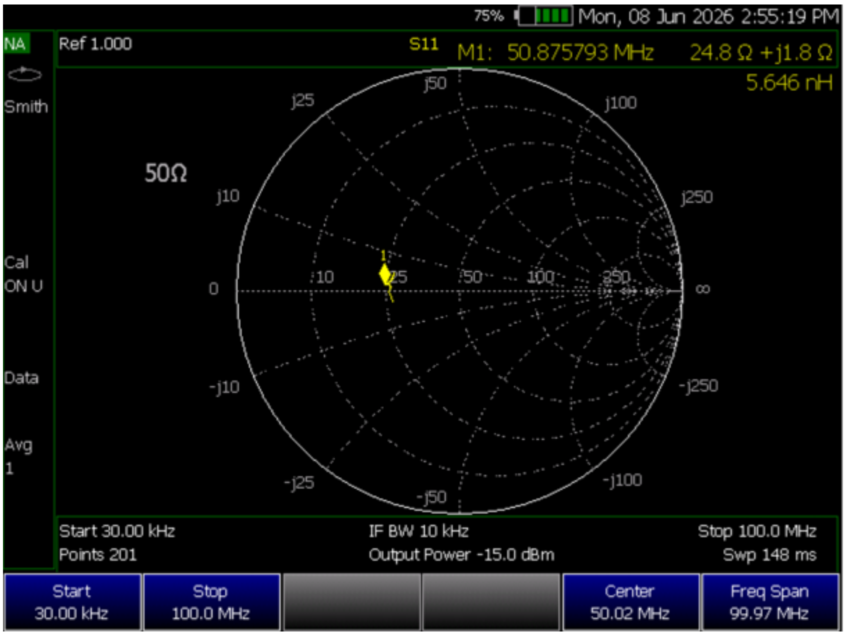

Figure 2.5: Smith chart measurement of the "Two by 2" termination setup, displaying an impedance of $24.8\,\Omega + j1.8\,\Omega$ at 50.875 MHz.

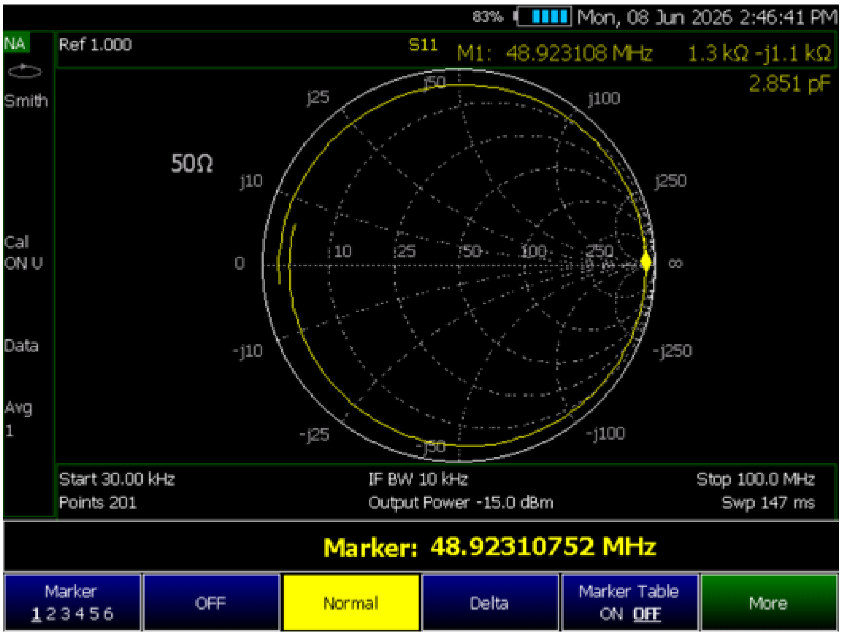

Figure 2.6: Smith chart measurement showing the profound impedance transformation ($1.3\,\mathrm{k\Omega} - j1.1\,\mathrm{k\Omega}$ at 48.923 MHz) when a short load is measured through an uncalibrated or specific length of connected coaxial cable.

2.1.6 Short discussion of measurement results

The measurements clearly illustrate the critical importance of proper VNA calibration and the behavior of parallel RF loads. Following calibration, the VNA accurately recorded the $75\,\Omega$ termination as $74.3\,\Omega$ with a negligible reactive component of $j0.3\,\Omega$, mapping correctly to the Smith chart's real axis. Similarly, the short circuit calibration was successful, yielding a near-ideal $-0.3\,\Omega$.
The experiment exploring makeshift terminations using the BNC Coax-Tee/Power Divider yielded excellent practical results. The "One by 2" termination measured approximately $50.2\,\Omega$, confirming a single standard load. When the "Two by 2" termination was measured, the impedance dropped to $24.8\,\Omega + j1.8\,\Omega$. This perfectly demonstrates the parallel resistance formula:

$$R_{\text{total}} = \frac{R_1 \times R_2}{R_1 + R_2} \tag{2.2}$$

where two $50\,\Omega$ loads connected in parallel via the tee result in an equivalent impedance of $25\,\Omega$.
Finally, Figure 2.5 highlights the impedance transformation effect of coaxial cables. When a short circuit is measured at the end of a transmission line without shifting the calibration pla to the end of the cable, the line acts as an impedance transformer, shifting the perceived impedance from $0\,\Omega$ to a highly complex value ($1.3\,\text{k}\Omega - j1.1\,\text{k}\Omega$) depending on the electrical length of the cable and the phase constant (β) at 48.923 MHz.

2.2 Task 3: Determine Radio Link Parameters

2.2.1 Task Description

The objective is to measure the return losses (S_{11}, S_{22}) and transmission losses (S_{21}, S_{12}) of a radio link using antennas.

2.2.2 Schematic

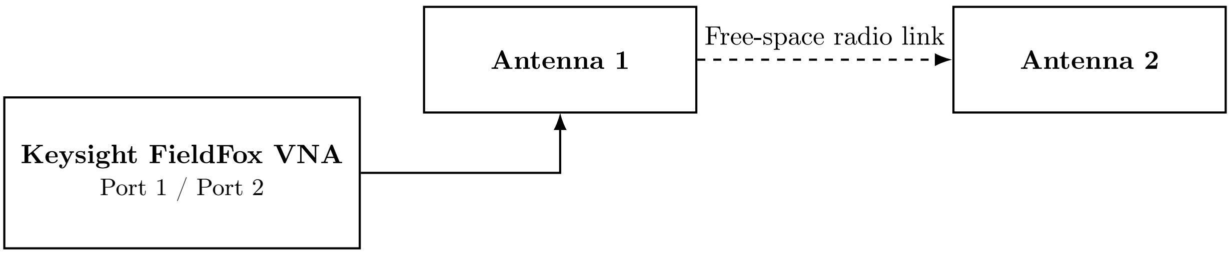

Figure 2.7: Measurement setup for radio link parameter analysis using antennas.

2.2.3 Formulas and Calculations

The VNA measures the S-parameters directly in decibels (dB), representing the logarithmic ratio of the reflected or transmitted wave to the incident wave.

$$\text{Return Loss (dB)} = -20\log_{10}|\rho| = -20\log_{10}|S_{11}| \quad \text{or} \quad -20\log_{10}|S_{22}| \tag{2.3}$$

$$\text{Transmission Loss (dB)} = -20\log_{10}|S_{21}| \quad \text{or} \quad -20\log_{10}|S_{12}| \tag{2.4}$$

2.2.4 Measurement results

The following tables summarize the S-parameter log-magnitude peak values extracted from the VNA plots for both the dual and single antenna setups.

Adjusted	Measured	
Frequency / MHz	**Parameter**	**Magnitude / dB**
972.335	S_{21}	-27.53
970.928	S_{22}	-26.55
989.511	S_{11}	-15.34
989.511	S_{21}	-72.27
989.511	S_{22}	-10.35

Table 2.2: Measured S-Parameters for the Dual Antenna Setup

Adjusted	Measured	
Frequency / MHz	**Parameter**	**Magnitude / dB**
260.132	S_{11}	-33.26
260.132	S_{12}	-26.99

Table 2.3: Measured S-Parameters for the One Antenna Setup

2.2.5 Curves & diagrams

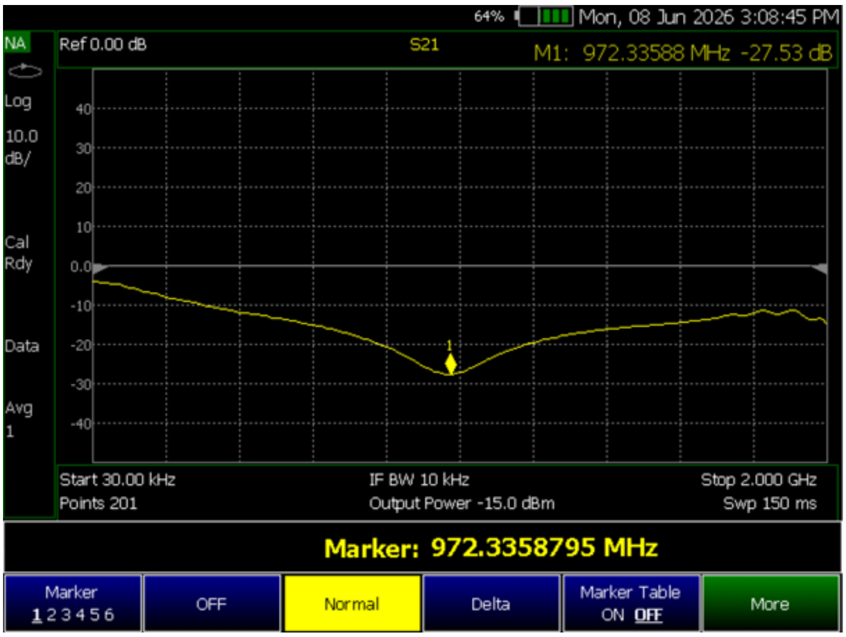

Figure 2.8: Log-magnitude plot of the transmission loss (S_{21}) for the dual antenna setup, showing a marker at 972.335 MHz with a value of -27.53 dB.

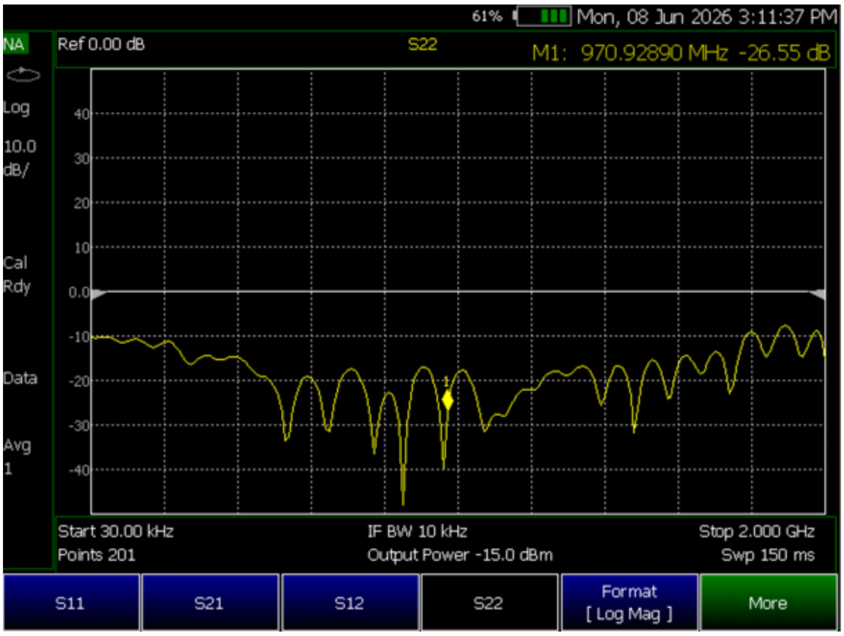

Figure 2.9: Log-magnitude plot of the return loss (S_{22}) for the dual antenna setup, showing a marker at 970.928 MHz with a value of -26.55 dB.

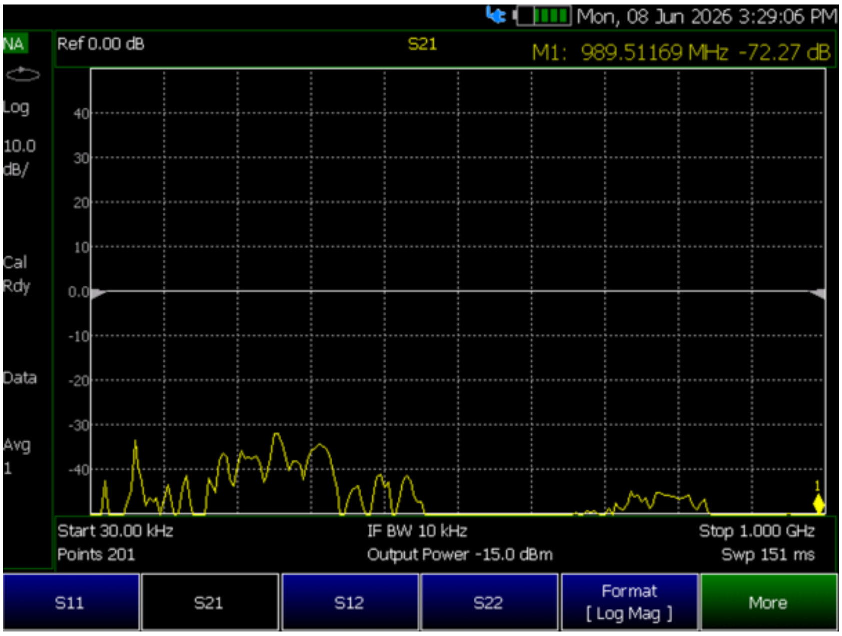

Figure 2.10: Log-magnitude plot of the transmission loss (S_{21}) for the dual antenna setup, showing a significant signal drop to -72.27 dB at 989.511 MHz.

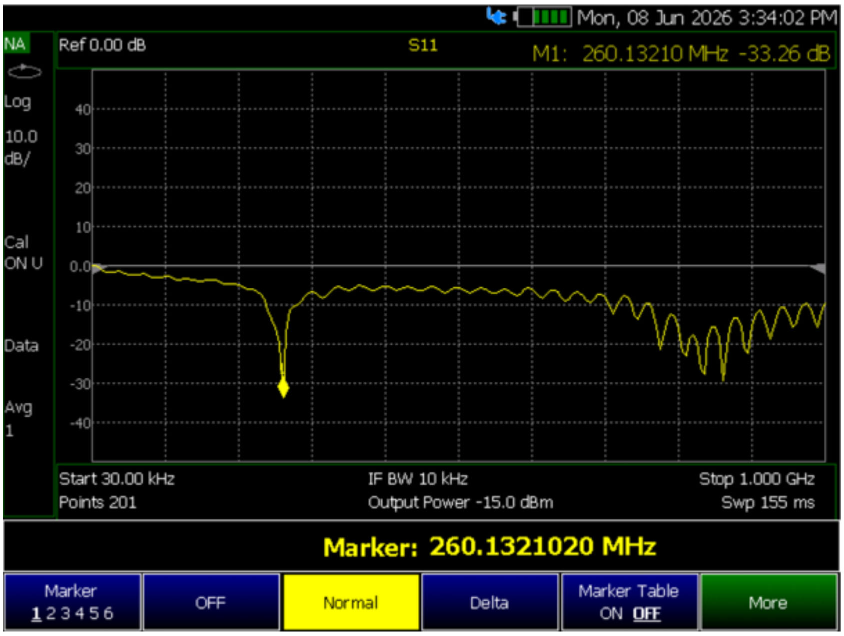

Figure 2.11: Log-magnitude plot of the return loss (S_{11}) for the single antenna setup, showing a distinct resonance peak of -33.26 dB at 260.132 MHz.

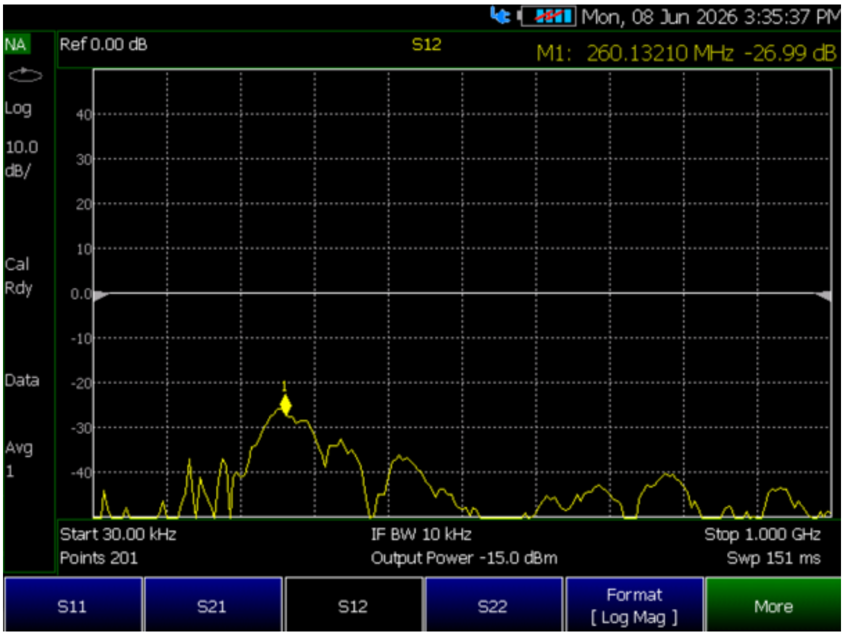

Figure 2.12: Log-magnitude plot of the transmission (S_{12}) for the single antenna setup, measured at -26.99 dB at 260.132 MHz.

2.2.6 Short discussion of measurement results

The measurements successfully document the return and transmission losses of the radio link. However, instead of the targeted 433 MHz ISM band, the antennas tested showed primary resonances at different frequencies.
For the dual antenna setup, a usable transmission loss (S_{21}) of -27.53 dB was recorded at 972.33 MHz, alongside a good return loss (S_{22}) of -26.55 dB at 970.92 MHz. At 989.51 MHz, the transmission loss drastically worsened to -72.27 dB, indicating very poor signal transfer at that specific frequency. The single antenna setup demonstrated an excellent return loss (S_{11}) of -33.26 dB at 260.13 MHz, meaning very little power was reflected back to the source at this frequency.

3 Instrument device list

Device	Manufacturer	Model	Serialnumber
Spectrum Analyser	Keysight	Fieldfox N9917A[1]	-
Voltage Controlled Oscillator	Mini-Circuits	ZOS-100[2]	-
Frequency Mixer	Mini-Circuits	ZX05-1-S+[3]	-

Table 3.1: Instrument device list

IMPORTANT NOTE: During the laboratory session for exercise 5, the Keysight Fieldfox spectrum analyser was facing issues and had to be rebooted, as well as soft and hard reset multiple times - to put it back into an operational state.
This event caused a significant delay due to which parts of the exercise, as well as aspects of the performed experiments were omitted.

Adarsh Anooj Graz, July 31, 2026

Juanith Mathew Thomas Graz, July 31, 2026

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References

[1] Fieldfox, Keysight, 2025, analyser datasheet. [Online]. Available: https://www.keysight.com/us/en/assets/7018-03314/data-sheets/5990-9783.pdf

[2] ZOS-100 Datasheet, Mini-Circuits, 2021, oscillator datasheet 1. [Online]. Available: https://www.minicircuits.com/pdfs/ZOS-100.pdf?srsltid=AfmBOophgqw6PHH4OXXISO9D2X6_8Kt0eZ8gK9uWvQuuj3A5ixO6vTHa

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